GAME DESIGN DOCUMENT

MyVT Gacha Collection

Browser-Based Idle Gacha Collection Game

VERSION
v3.1 - Monetisation Edition

LAST UPDATED
June 2025

ROSTER
319 Malaysian VTubers

PLATFORM
Browser (GitHub Pages)

5 currencies | 4 studio stations | 10 studio levels | 1 minigame | 17 sections

Built with vanilla HTML, CSS, and JavaScript
No frameworks. No backend. Zero install.
Data sourced from hololist.net

GDD v3.1

Table of Contents

1. Game Overview

MyVT Gacha Collection is a browser-based idle gacha collection game themed around the Malaysian VTuber (MyVT) community. Players collect virtual cards featuring real Malaysian VTubers by pulling from gacha banners, then manage a virtual studio where assigned VTubers generate resources passively. Built entirely with vanilla HTML, CSS, and JavaScript, the game is hosted on GitHub Pages and requires no installation, no backend server, and no database. All game state is persisted using the browser's localStorage, making it a truly zero-friction experience that players can enjoy on any device with a modern browser.

The roster features 319 Malaysian VTubers sourced from hololist.net, spanning independent creators and members of various agencies including VGakuenLive, HoloDream, MyHolo TV (VILIT), Projek Hikayat, and Phase Connect. Each VTuber is represented by a character card with portrait art from their hololist entry, and cards can be pulled in three variant tiers: Normal, SR (Super Rare), and SSR (Super Super Rare). The variant tier determines the card's visual frame, resource output multiplier, and level cap, creating a compelling collection and progression system built around the diverse Malaysian VTuber community.

The game design is guided by five core pillars: Accessibility, meaning zero install and any-browser compatibility; Community Celebration, showcasing the vibrant MyVT scene to a wider audience; Idle-Friendliness, where progress continues even when the player is offline; Collection Satisfaction, with 319 characters and variant upgrades providing long-term goals; and Skill Development, as the project serves as a practical learning exercise for web game development using vanilla technologies.

The primary target audience consists of members of the MyVT community, including VTubers themselves, their fans, and supporters of the Malaysian VTuber scene. Secondary audience members include general gacha game enthusiasts and idle game players who may discover the game through community sharing. The game is designed to be approachable for newcomers while offering enough depth to retain experienced gacha players. All text in the game is presented in English, reflecting the multilingual nature of the Malaysian VTuber community.

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2. Core Game Loop

The core gameplay loop of MyVT Gacha Collection revolves around a seven-step cycle that intertwines gacha pulling, idle resource generation, and character progression. Players begin by spending Stars on gacha pulls to acquire VTuber character cards. These cards are then assigned to Studio Room stations, where they passively generate resources even when the player is offline. Accumulated resources are reinvested to level up characters, ascend them to higher variant tiers, upgrade studio stations, and ultimately unlock new features and stations. The loop is designed to be simple enough to understand in seconds but deep enough to sustain engagement over weeks and months of regular play, with each step feeding naturally into the next.

New players start with 1,000 Stars, enough for approximately 10 single pulls on the Standard Banner. The first few pulls will typically yield Normal variant characters, which the player can immediately assign to the Stream Room station to begin generating Stars passively. This establishes the resource generation foundation that fuels the rest of the gameplay loop. As players progress, they unlock additional station types, each producing different currencies required for character progression, creating a web of interdependencies that keeps all game systems relevant throughout the play experience.

Step	Action	Description
1	Pull	Spend Stars on gacha banners to acquire VTuber character cards.
2	Assign	Place collected characters into Studio Room stations to generate resources.
3	Earn	Resources accumulate passively (idle), even while the player is offline.
4	Level Up	Spend Star Dust and Stars to increase character levels, boosting station output.
5	Ascend	Spend Star Fragments to upgrade Normal to SR and SR to SSR variants.
6	Expand	Upgrade Studio Level to unlock new stations, slots, and features.
7	Repeat	Continue pulling to fill collection gaps; aim for 100% SSR completion.

Table 1: The Seven-Step Core Game Loop

3. Gacha Pull System

The gacha pull system is the primary mechanism through which players acquire new character cards. It features two banner types, three variant tiers with distinct drop rates, a pity system to prevent excessive bad luck, and a duplicate handling system that ensures every pull produces meaningful value. The system is designed to create excitement and anticipation with each pull while maintaining fairness and long-term progression for dedicated players.

3.1 Banner Types

The gacha system features two types of banners. The Standard Banner contains all 319 Malaysian VTubers from the hololist roster and is permanently available, serving as the primary source of new characters. The Featured Banner is a rotating limited-time banner that highlights 3-4 specific VTubers with increased drop rates on a weekly rotation. Both banners draw from the full 319-character pool for character selection, meaning that any character can be pulled from either banner. The Featured Banner's rate-up characters simply have a higher probability of appearing, making it the preferred choice for players seeking specific VTubers for their collection.

Banner Type	Pool Size	Featured Characters	Rotation
Standard	319	None (equal rates)	Permanent
Featured	319	3-4 rate-up VTubers	Weekly

Table 2: Banner Types Overview

3.2 Pull Rates

Each pull yields one character card with a randomly determined variant tier. The variant system does not affect which character is obtained; rather, it determines the rarity and visual quality of that character's card. Normal cards appear at a base rate of 82%, SR cards at 15%, and SSR cards at 3%. This means that on average, a player will see an SSR card roughly once every 33 pulls, though the pity system ensures that no player goes more than 50 pulls without an SSR. The 10x multi-pull additionally guarantees at least one SR or above card, providing a reliable floor for multi-pull investments.

Variant	Drop Rate	Card Frame	Pull Guarantee
Normal	82%	Grey border, standard art	Always obtainable
SR	15%	Silver border with glow	1 per 10x pull
SSR	3%	Gold border with shine	Pity at 50 pulls

Table 3: Pull Rates by Variant Tier

3.3 Pull Costs

Pulling from either banner costs Stars, the game's primary currency. A single pull costs 100 Stars, while a ten-pull costs 1,000 Stars (equivalent to 10 single pulls with no discount). The pricing is intentionally simple and consistent across both banner types, ensuring that players can easily calculate how many pulls they can afford at any given time. The starting bonus of 1,000 Stars provides enough for either 10 single pulls or one multi-pull, giving new players an immediate and meaningful introduction to the gacha mechanic.

Pull Type	Cost (Stars)	Guarantee
Single Pull (1x)	100	None
Multi Pull (10x)	1,000	At least 1 SR or above

Table 4: Pull Costs

3.4 Pity System

A pity system ensures that players are guaranteed an SSR variant after 50 consecutive pulls without obtaining one. The pity counter tracks pulls across all banners and resets every time an SSR variant is pulled, regardless of which banner it came from. This cross-banner tracking means that players cannot accidentally lose their pity progress by switching banners, and it prevents excessively unlucky streaks from undermining the player experience. Additionally, the 10x multi-pull guarantees at least one SR or above card, providing a reliable floor for multi-pull investments and ensuring that even unlucky multi-pulls yield at least one notable result.

3.5 Duplicate Handling

When a player pulls a character they already own at the same variant tier, the duplicate is converted into Shards. Shards serve as a universal resource that can be exchanged for Star Dust or Star Fragments at a conversion rate displayed in the in-game interface. If a player already owns a character at SSR tier and pulls that same character at any tier again, the duplicate is automatically converted into bonus Stars instead of Shards, providing a meaningful endgame resource sink. This duplicate handling system ensures that every pull produces value, even when the character pool has been largely exhausted, which is especially important given the relatively small roster of 319 characters.

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4. Studio Room Management

The Studio Room is the management hub of MyVT Gacha Collection, providing the idle resource generation layer that transforms the game from a simple pull simulator into an engaging management experience. Players assign their collected VTuber characters to various Stations, each of which produces a different type of resource. The strategic decision of which characters to assign to which stations, combined with station upgrades and character progression, creates a deep and satisfying management loop that rewards thoughtful planning and long-term engagement.

4.1 Station Types

Each station represents a different activity that VTubers might engage in, and each produces a unique resource. Players start with the Stream Room station unlocked at Studio Level 1 and gradually unlock additional stations by increasing their Studio Level. Every station can hold exactly one assigned VTuber at a time, and the player may reassign characters freely with no cooldown or penalty. The output of each station scales with the assigned character's variant tier and level, as well as the station's own upgrade level, creating multiple axes of optimization for players to explore.

Station	Resource	Primary Use	Unlock
Stream Room	Stars	Gacha pulls, station upgrades	Studio Lv 1
Creative Corner	Star Dust	Character levelling	Studio Lv 2
Practice Hall	Star Fragments	Ascension material	Studio Lv 4
Lounge	Bond Points	Studio Level EXP	Studio Lv 6

Table 5: Station Types and Their Resources

4.2 Station Upgrades

Each station can be upgraded from Level 1 through Level 5. Upgrading increases the station's output multiplier, meaning that the same assigned character will produce more resources per minute at higher station levels. The output multiplier progresses as follows: Level 1 provides a x1 multiplier, Level 2 provides x1.5, Level 3 provides x2, Level 4 provides x3, and Level 5 provides x5. This exponential scaling means that maxing out a station is a significant long-term investment that dramatically increases resource generation, and the scaling costs ensure that players must carefully prioritise which stations to upgrade first.

Level	Multiplier	Cost (Stars)	Cost (Fragments)
1	x1	(Base)	(Base)
2	x1.5	500	50

3	x2	2,000	200
4	x3	8,000	800
5	x5	30,000	3,000

Table 6: Station Upgrade Costs and Multipliers

4.3 Resource Output Calculation

The resource output per minute for any station is determined by a multiplicative formula: Output = Base Rate x Character Variant Multiplier x Character Level Bonus x Station Level Multiplier. The base rate differs per station type. Character variant multipliers are x1 for Normal, x2 for SR, and x5 for SSR. The character level bonus is calculated as 1 + (Level x 0.02), meaning a Level 50 SSR character at a Level 5 station produces (Base Rate x 5 x 2 x 5) = 50x the base rate. This formula ensures that every aspect of character progression meaningfully contributes to resource generation, making both pulling higher-tier cards and investing in levelling feel rewarding.

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5. Currency System

MyVT Gacha Collection features five distinct currencies, each serving a unique purpose within the game economy. This multi-currency design ensures that players must engage with all aspects of the game to progress, preventing the common gacha game problem of a single currency becoming irrelevant after the early game. Four of the five currencies are produced by studio stations, while the fifth, Gems, is a premium currency earned primarily through the Super Chat Toss minigame and collection milestones. Each currency feeds into different progression systems, creating interdependencies that make the gameplay loop feel cohesive and rewarding throughout the entire play experience.

Currency	Symbol	Source	Primary Use
Stars	*	Stream Room, Daily Login, Duplicate SSR	Gacha pulls, Station upgrades
Star Dust	+	Creative Corner, Shard conversion	Character levelling
Star Fragments	#	Practice Hall, Shard conversion	Ascension crafting

Bond Points	~	Lounge station, Odekake outings	Studio Level EXP
Gems	$	Minigame, Milestones	Premium rewards, exchange

Table 7: Currency Types, Sources, and Uses

New players receive a starting package of 1,000 Stars to immediately begin pulling on banners. No other currencies are provided at the start; players must progress through the early game to unlock stations and begin generating secondary currencies. The Daily Login Bonus provides 100 Stars as a base amount, increasing by 50 Stars per consecutive day of login, with a daily cap of 500 Stars after a 9-day streak. Missing a day resets the streak to day 1, creating a gentle but meaningful incentive for regular engagement. Gems, the premium currency, are earned by playing the Super Chat Toss minigame and by reaching collection milestones, providing an additional progression path beyond passive resource generation.

6. Character Progression

Characters in MyVT Gacha Collection progress through two interconnected systems: levelling and ascension. Levelling strengthens a character within its current variant tier, increasing its resource output when assigned to a station. Ascension upgrades a character to a higher variant tier (Normal to SR, or SR to SSR), unlocking a new card frame, higher level caps, and significantly increased output multipliers. Both systems require different currencies, ensuring that players must engage with multiple station types to fully develop their roster. The progression system is designed to provide constant short-term goals while maintaining compelling long-term targets for dedicated players.

6.1 Character Levelling

Each character can be levelled from Level 1 up to a maximum level determined by their current variant tier. Normal variant characters have a level cap of 20, SR characters can reach Level 35, and SSR characters can reach Level 50. Levelling requires spending Star Dust and Stars, with costs scaling based on the target level. Each level increases the character's resource output by approximately 2% (the level bonus formula is 1 + Level x 0.02), making levelling a consistent and predictable investment that pays dividends across all station types where the character might be assigned.

Variant	Level Cap	Cost (Star Dust)	Cost (Stars)

Normal	20	10 + (Lvl x 5)	20 + (Lvl x 10)
SR	35	30 + (Lvl x 10)	50 + (Lvl x 20)
SSR	50	80 + (Lvl x 20)	100 + (Lvl x 40)

Table 8: Levelling Costs by Variant Tier

6.2 Character Ascension

Ascension is the process of upgrading a character to a higher variant tier. To ascend a Normal character to SR, the character must be at maximum level (20) and the player must spend 100 Star Fragments. To ascend an SR character to SSR, the character must be at its maximum SR level (35) and the player must spend 500 Star Fragments. Ascension changes the character's card frame, increases the variant output multiplier (from x1 to x2 for Normal-to-SR, and from x2 to x5 for SR-to-SSR), raises the level cap, and increases Studio EXP contribution. This dual-gate system (max level + fragment cost) ensures that ascension is a significant milestone that requires sustained investment in each individual character.

Ascension	Required Level	Cost (Fragments)	New Level Cap
Normal to SR	Lv 20	100	35
SR to SSR	Lv 35	500	50

Table 9: Ascension Requirements

7. Studio Level and Feature Unlocks

The Studio Level serves as the overarching progression gate for the game. Unlike character levels or station upgrades, the Studio Level does not directly improve resource generation rates. Instead, it unlocks new features, stations, and station slots, providing a structured sense of progression that guides players through the game's systems at a measured pace. Studio EXP is earned passively from assigned characters in stations, with SSR characters contributing more EXP than SR, and SR contributing more than Normal. The Studio Level ensures that new players are not overwhelmed by too many systems at once while providing experienced players with clear goals to work toward over the course of their playthrough.

Studio Lv	Unlocks	EXP Required

1	Stream Room + 2 station slots	0
2	Creative Corner station	200
3	3rd station slot	600
4	Practice Hall station	1,500
5	4th station slot	3,500
6	Lounge station	7,000
7	Character detail stats	12,000
8	5th station slot	20,000
9	Agency filter	35,000
10	Odekake (Going Out)	60,000

Table 10: Studio Level Progression and Unlocks

7.1 Odekake (Going Out) - Planned Feature

At Studio Level 10, players unlock the Odekake feature, inspired by the training camp system in Uma Musume. Odekake allows players to send a character on a timed "outing" lasting between 1 and 4 hours. During this time, the character cannot be assigned to any station. When the outing concludes, the character returns with Bond Points, Studio EXP, and occasionally rare crafting materials. Repeated outings with the same character build a Bond Level, which unlocks character-specific bonuses such as increased output multipliers or reduced ascension costs. This feature adds an emotional attachment layer to the game, encouraging players to invest in their favourite characters beyond mere numerical optimization. The Bond Level system provides a long-term progression goal that extends beyond the collection and variant upgrade mechanics, giving players a reason to continue engaging with the game even after achieving a high collection completion percentage.

8. Collection Gallery

The Collection Gallery is the player's visual archive of all 319 Malaysian VTubers in the game. It displays every character as a card in a responsive grid layout, with distinct visual indicators showing which characters the player owns, which variant tiers they have obtained, and which characters remain undiscovered. The gallery serves as both a progress tracker and a showcase of the player's collection achievements, providing a clear visual representation of how far the player has come and what remains to be collected.

The gallery provides multiple filter options to help players navigate the large roster. Ownership filters allow viewing All, Owned, or Unowned characters. Variant filters narrow results by highest owned variant (Normal, SR, or SSR). An Agency filter, unlocked at Studio Level 9, enables filtering by agency affiliation such as Independent, VGakuenLive, VILIT, and others. A Station filter shows characters by assignment status (Any, Currently Assigned, or Unassigned). Players can also search by VTuber name and sort by Name or Variant tier. The gallery displays 40 characters per page with pagination controls.

A collection progress bar at the top of the gallery shows overall completion percentage alongside per-variant completion stats (for example, 45/319 SSR, 120/319 SR, 280/319 Normal). Collection milestones at 10, 25, 50, 100, 150, 200, 250, 300, and 319 owned characters reward players with bonus resources and Gems, providing structured goals and celebration moments throughout the collection journey. Clicking any character card opens a detail popup showing the character's pull history, current stats, all owned variants, and variant-specific information.

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9. UI/UX Design

The game's user interface is built around a dark theme that follows the visual language common in anime-style gacha games, creating an immersive and familiar aesthetic for the target audience. The primary background is a deep dark grey (#0e0e0c), with card backgrounds in slightly lighter tones to create visual depth. Purple accents (#896fd9) are used for interactive elements and headers, teal/green accents (#5dd499) highlight positive feedback such as resource gains, and gold tones are reserved for SSR-tier elements to convey rarity and premium quality.

9.1 Navigation

The user interface employs a tab-based navigation system with five primary sections: Home, Pull, Collection, Studio, and Minigame. A persistent top navigation bar displays these tabs along with the player's current resource counts (Stars, Star Dust, Star Fragments, Bond Points) and a stamina bar. A settings overlay accessible from the navigation bar provides save, export, import, device transfer, and reset functionality. The navigation is designed to be thumb-friendly on mobile devices, with tabs positioned at the bottom on narrow viewports and at the top on wider screens.

Tab	Content	Key Interactions

Home	Stats, daily login, offline earnings, studio overview	Claim offline earnings, view progress
Pull	Banner selection, pull animation, results	Single/multi pull, banner switching
Collection	Character gallery grid with filters	Filter, sort, view character details
Studio	Stations, assignment, upgrades	Assign characters, upgrade stations
Minigame	Super Chat Toss game screen	Play rounds, view high scores

Table 11: Navigation Tabs and Their Content

9.2 Card Design

Character cards are the central visual element of the game. Each card displays the VTuber's portrait image from hololist.net, their name, agency, and variant tier badge. Normal cards have a simple grey border, SR cards feature a silver border with a subtle glow effect, and SSR cards display a gold border with an animated shine effect. Cards are sized at approximately 120x160 pixels in the gallery grid and scale up to 200x280 pixels in single-pull reveal and 120x168 pixels in 10-pull reveal. The pull animation features a sequential slide-in with a 3D flip reveal, where cards appear one at a time from below the screen and flip to reveal the character. Single pulls take approximately 1 second for a dramatic reveal, while 10-pulls complete in about 8 seconds total with an option to tap and skip.

9.3 Pull Animation

The pull animation system employs a sequential slide-in combined with a 3D card flip. Cards appear one at a time from below the viewport, slide up to the centre, and flip to reveal the character. For single pulls, this takes approximately 1 second with a dramatic pause before the flip. For 10-pulls, the total animation runs for about 8 seconds with cards appearing in rapid succession. Players can tap to skip the animation at any point. SR pulls trigger a silver screen flash effect, and SSR pulls trigger a golden screen flash accompanied by an animated golden glow ring around the card. These visual effects create moments of excitement and celebration that reinforce the emotional reward of pulling higher-tier cards.

9.4 Responsive Design

The game is designed to be fully responsive, adapting to screen sizes from 320px mobile screens to 1920px desktop displays. On mobile, the gallery grid displays 3 cards per row with a bottom navigation bar. On tablet, the grid expands to 4-5 cards per row. On desktop, up to 8 cards per row are shown with a horizontal top navigation bar. The Studio Room layout adapts similarly, showing stations in a vertical stack on mobile and a grid on desktop. All interactive elements have touch targets of at least 44x44 pixels for comfortable mobile interaction.

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10. Super Chat Toss Minigame

The Super Chat Toss is a tap-based 30-second minigame accessible from the Minigame navigation tab. It provides an active gameplay alternative to the idle resource generation of the studio system, offering players a skill-based way to earn Stars and Gems. The minigame features five bubble types that float across the screen: Coin bubbles grant +10 Stars, Premium bubbles grant +25 Stars, Gem bubbles grant +1 Gem, Landmine bubbles penalise the player with -15% of their current score, and Gift bubbles add +3 seconds to the remaining time, extending the play session.

A boost meter fills as the player taps bubbles. When the meter reaches 100%, it triggers Super Chat Mode for 5 seconds, during which all Coin and Premium bubbles are worth 5 times their normal value. This creates exciting moments of high-scoring potential that reward sustained accuracy and quick reflexes. Before each round, the player selects an owned VTuber as their "streamer lead," whose variant tier determines a score multiplier: Normal characters provide x1, SR characters provide x1.5, and SSR characters provide x2. This character lead system adds a strategic layer, encouraging players to invest in and use their highest-tier characters during minigame rounds.

Each round of Super Chat Toss costs 15 stamina points. The results screen at the end of each round displays the Stars earned, Gems earned, and Bond Points gained (+10 per round). High scores are tracked locally, giving players a personal best to beat. The minigame serves multiple design purposes: it provides active engagement for players who want more than idle gameplay, it creates a meaningful use for stamina, it offers a path to earn the premium Gem currency, and it gives owned characters additional utility beyond studio station assignment.

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11. Stamina System

The stamina system governs access to the Super Chat Toss minigame, adding a resource management layer to the active gameplay experience. Players have a maximum of 200 stamina points, which regenerate at a rate of 1 point every 4 minutes (approximately 360 points per day at full regeneration). Each round of Super Chat Toss costs 15 stamina, meaning a fully charged stamina bar allows for approximately 13 consecutive rounds of play before the player must wait for regeneration.

The stamina bar is prominently displayed in both the navigation bar and the minigame start screen, alongside a timer showing the time until the next point of recovery. When the player loads the game after being offline, the system calculates offline recovery based on the elapsed time since the last save, ensuring that players do not lose any regeneration progress during breaks. This offline recovery calculation is capped to prevent excessive accumulation during very long absences. The stamina system creates a natural cadence for minigame engagement, preventing the minigame from becoming the sole focus of gameplay while still providing regular opportunities for active play sessions.

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12. Firebase Cloud Save

Firebase Cloud Save is a planned feature that will allow players to synchronise their game progress across multiple devices by leveraging Firebase Authentication and Cloud Firestore as a cloud backend. This represents the first significant departure from the game's purely client-side architecture, introducing a lightweight server-side component to enable cross-device save data portability. The feature is designed to remain entirely optional; players who prefer the current localStorage-only approach can continue using export/import codes without ever creating a cloud account. The primary motivation for this feature is the significant number of player support requests related to lost save data due to browser cache clearing, device replacement, or accidental resets.

12.1 Overview

The Firebase Cloud Save system combines Firebase Authentication for user identity management with Cloud Firestore for persistent save data storage. Players authenticate once through an anonymous account created automatically on first launch, and can optionally upgrade to a persistent account using Google or Discord sign-in. All game state data is serialised to JSON and stored in a Firestore document keyed by the player's authenticated user ID (UID). The system supports seamless save/load across devices, automatic background synchronisation, and manual save/load triggers accessible from the settings menu. This approach was chosen over a custom backend because Firebase's free tier is generous enough to support the expected player base without requiring server infrastructure or ongoing maintenance costs.

12.2 Architecture

The cloud save architecture follows a client-initiated sync pattern. On the client side, a new module (js/firebase.js) will manage all Firebase interactions, including authentication state changes, Firestore read/write operations, and conflict detection. The authentication flow begins with an anonymous sign-in using Firebase Anonymous Auth, which creates a unique UID without requiring any user input. Players can later link their anonymous account to a Google or Discord identity through the settings menu, which preserves all existing save data while upgrading to a recoverable account. On the server side, Firestore uses a collection named "saves" with documents keyed by UID, where each document contains the complete game state JSON along with metadata fields for version tracking and timestamp comparison.

Component	Technology	Purpose
Auth - Anonymous	Firebase Anonymous Auth	Auto-create account on first launch, no user input needed
Auth - Persistent	Google / Discord OAuth	Optional upgrade for account recovery
Database	Cloud Firestore	Store serialised game state per user
Client Module	js/firebase.js	Auth state, sync logic, conflict resolution
Storage Key	saves/{uid}	One document per user containing full game state

Table 12: Firebase Cloud Save Architecture

12.3 Data Model

The Firestore document for each user contains two top-level sections: a metadata object and a gameData object. The metadata includes a "lastModified" timestamp (ISO 8601), a "schemaVersion" string matching the current save format version, and a "deviceFingerprint" string to identify which device last wrote the data. The gameData object contains the exact same JSON structure as the localStorage save, including all currency balances, character collection data, studio configuration, gacha pity state, stamina timers, milestone tracking, pull history, and daily login streak. Key fields synced include stars, starDust, starFragments, bondPoints, gems, the characters collection object, studio level and assignments, gacha pityCounter, stamina state, and all milestone claim records.

12.4 Sync Strategy

The synchronisation strategy employs a last-write-wins conflict resolution model based on the lastModified timestamp. Auto-save triggers every 60 seconds when the player is online and authenticated, uploading the current game state to Firestore if the local lastModified timestamp is newer than the remote timestamp. On game load, the system fetches the remote save and compares timestamps: if the remote save is newer, it merges remote data into the local state. A manual "Cloud Save" and "Cloud Load" button pair in the settings menu allows players to force an immediate upload or download at any time. The sync system is designed to be resilient to network failures, queuing failed writes and retrying on the next successful connection without blocking gameplay.

12.5 Firebase Free Tier Analysis

Firebase's Spark (free) tier provides generous limits that are expected to cover the game's needs during its early lifecycle. The Firestore free tier includes 50,000 reads per day, 20,000 writes per day, 1 GiB of stored data, and unlimited authentication operations. For the MyVT Gacha Collection player base, each active player generates approximately 1 write per 60-second auto-save interval (approximately 240 writes per 4-hour session) and 1 read on game load, plus 1 read per sync check. This means the free tier can comfortably support roughly 80-100 daily active players on auto-save alone, with significantly higher capacity if auto-save intervals are increased to 120 or 180 seconds.

Player Base (DAU)	Est. Daily Writes	Est. Daily Reads	Free Tier?	Est. Monthly Cost
100	24,000	1,000	Yes	$0
1,000	240,000	10,000	No	~$3-5
10,000	2,400,000	100,000	No	~$30-50
100,000	24,000,000	1,000,000	No	~$300-500

Table 13: Firebase Cost Estimates by Player Base Size

12.6 Security Rules

Firestore security rules will enforce strict per-user data isolation to prevent unauthorised access to player save data. The rules allow authenticated users to read and write only their own document in the "saves" collection, verified by matching the document ID to the request.auth.uid field. Write operations will additionally validate the data structure by checking that required top-level fields (gameData, lastModified, schemaVersion) are present in the incoming document. This prevents malformed data from corrupting saves, although the validation is intentionally lightweight to avoid complex rule evaluation costs. More detailed validation will be handled on the client side before any write attempt, ensuring that only well-formed game state is uploaded to Firestore.

12.7 Privacy Considerations

The Firebase Cloud Save system is designed with privacy as a core concern. No personally identifiable information (PII) is collected beyond what is provided by the authentication provider: for anonymous auth, only a random UID is generated with no personal data; for Google auth, the player's email and display name are accessible but only the UID is stored in the save document; for Discord auth, the Discord user ID and username are available but again only the UID is stored. Game analytics such as play time, session frequency, and collection progress are not tracked by the cloud save system, maintaining the game's minimal data footprint. A future consideration under GDPR compliance is whether a data deletion endpoint should be exposed, allowing players to request complete removal of their cloud save data. For the initial implementation, players can effectively delete their data by resetting the in-game save from the settings menu, which triggers a Firestore document deletion.

12.8 Sprint 9 Implementation Plan

The Firebase Cloud Save feature is planned for Sprint 9, with an estimated timeline of 3-4 weeks for a solo developer. The implementation is divided into four phases: Phase 1 (Week 1) covers Firebase project setup, service account configuration, and the anonymous auth flow with automatic account creation on first launch. Phase 2 (Week 2) implements the Firestore read/write layer, including the data serialisation/deserialisation logic and the last-write-wins conflict resolution algorithm. Phase 3 (Week 3) adds the Google and Discord OAuth sign-in options with account linking for existing anonymous accounts, along with the settings menu UI for cloud save/load controls and account management. Phase 4 (Week 4) is dedicated to testing, edge case handling (offline scenarios, simultaneous multi-device conflicts, large save data), and security rule refinement.

Phase	Week	Deliverables	Status
1	Week 1	Firebase project setup, anonymous auth flow	Not Started
2	Week 2	Firestore read/write, conflict resolution	Not Started
3	Week 3	OAuth sign-in, settings UI, account linking	Not Started
4	Week 4	Testing, edge cases, security rules	Not Started

Table 14: Sprint 9 Implementation Phases

13. Technical Architecture

The game is built entirely with vanilla web technologies: HTML5 for structure, CSS3 for styling and animations, and JavaScript ES6+ for game logic. No frameworks, libraries, or build tools are required, making the project accessible to developers of all skill levels and easy to deploy on any static hosting platform. The only external dependency is the character portrait images hosted on hololist.net, which are loaded dynamically via URLs stored in the game's character data file. This zero-dependency approach was chosen deliberately to keep the project simple, educational, and easy to maintain.

13.1 Technology Stack

Technology	Purpose	Rationale
HTML5	Page structure	Universal browser support
CSS3	Styling, animations, responsive layout	Card flip, dark theme, media queries
JavaScript ES6+	Game logic, state, DOM	No framework overhead
localStorage	Client-side persistence	No backend, instant save/load
JSON	Character roster data	Human-readable, easy updates

Table 15: Technology Stack

13.2 File Structure

The project follows a clean, modular file structure that separates concerns for maintainability and readability. Each JavaScript file handles a distinct area of game functionality, and the character data is stored in a separate JSON file that can be updated independently. Note that the original GDD listed js/studio.js and js/firebase.js, which do not exist in the current codebase. Studio logic is handled within game.js, and Firebase cloud save has not been implemented beyond placeholder UI buttons in the settings modal.

File	Purpose
index.html	Main page, loads all scripts
css/style.css	All styles: dark theme, animations, responsive
js/data.js	Character data loader (fetches characters.json)

js/game.js	Core state, save/load, tick, currencies, stamina, studio, milestones
js/gacha.js	Pull logic, rates, pity, featured banner, Fisher-Yates shuffle
js/characters.js	Levelling, ascension, shards, variant management
js/minigame.js	Super Chat Toss engine (bubbles, boost, scoring)
js/ui.js	UI rendering, navigation, pull animation, gallery, studio UI, toasts
data/characters.json	319 VTuber entries (name, slug, image, agency, url)

Table 16: Project File Structure

13.3 Save/Load System

All game state is persisted to the browser's localStorage under a single JSON key. The game auto-saves every 30 seconds and on every significant player action, ensuring minimal data loss. Players can also manually save via a button in the settings overlay. Export and import functionality allows players to transfer save data between devices using Base64-encoded save codes. A migration system handles save version updates, automatically converting old save data to the latest format when the game loads. Error resilience is built into the initialisation process: a try-catch block on startup catches any corruption or compatibility issues and displays an error message directing the player to check the browser console, rather than showing an infinite "Loading..." screen.

13.4 Game Tick System

A 1-second interval game tick runs whenever the game is open, updating resource generation, stamina recovery, and the user interface. When the player closes the game and returns later, offline earnings are calculated based on the elapsed time since the last save, with a cap of 12 hours of offline resource accumulation. This ensures that extended offline periods do not result in excessively large resource windfalls that could unbalance the game economy. The tick system also handles cooldown timers, UI refreshes, and any time-dependent game events.

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14. Development Status and Sprint Plan

The original 9-sprint development plan has been partially completed with significant scope changes. The first four sprints (Core UI Shell, Gacha Pull System, Collection Gallery, and Studio Room) have been fully completed, establishing the core gameplay loop. Sprint 5 was reimplemented as Profile and History (originally planned as Currency and Idle System, but most of that functionality was integrated into earlier sprints). Sprint 8 (Super Chat Toss) was pulled forward and completed, displacing the originally planned Polish and Save/Load sprint. The remaining sprints show varying levels of completion.

Sprint	Focus	Status
1	Core UI Shell	Completed
2	Gacha Pull System	Completed
3	Collection Gallery	Completed
4	Studio Room	Completed
5	Profile and History	Completed / Partial
6	Character Progression	Partial
7	Studio Progression	Partial
8	Super Chat Toss	Completed
9	Cloud Save Sync	Not Started

Table 17: Sprint Status Overview

Several code review items have been deferred to future sprints. High-priority deferred items include XSS vulnerabilities in user-facing text (HP-03) and deprecated API usage (HP-05). Medium-priority items include an auto-backup system (V4-08). Low-priority polish items (LP-01 through LP-05) include various UI refinements and edge-case handling improvements. These items are documented in the code review tracker and will be addressed in a dedicated code quality sprint.

Proposed future sprints include Sprint 10 (Code Quality and Security), Sprint 11 (Odekake Going Out feature requiring Studio Level 10), Sprint 12 (Social Features including leaderboards and friend lists), and Sprint 13 (Content Updates with new VTuber additions, seasonal events, and limited-time banners). These sprints represent the long-term development roadmap beyond the initial release, extending the game's content and feature set based on player feedback and community engagement.

15. Character Data

The character roster is sourced from hololist.net, which maintains a comprehensive database of Malaysian VTubers. Each character entry in the data file (characters.json) contains five fields: name, slug, image URL, agency, and profile URL. The slug serves as a unique identifier used for internal game logic and save data references, while the image URL points to the character's portrait on hololist.net. The agency field is used for filtering in the Collection Gallery (unlocked at Studio Level 9) and for display on character cards.

15.1 Agency Distribution

The 319-character roster is heavily weighted toward independent creators, who make up approximately 79.6% of the total roster. This distribution reflects the current state of the Malaysian VTuber scene, where independent creators form the overwhelming majority. The largest agency represented is VGakuenLive with 7 members (2.2%), followed by HoloDream with 6 members (1.9%), Slice of Six Crew with 5 members (1.6%), and AmaiUsagi with 4 members (1.3%). The remaining 43 characters (13.5%) belong to various smaller agencies and groups.

Agency	Count	Percentage
Independent	254	79.6%
VGakuenLive	7	2.2%
HoloDream	6	1.9%
Slice of Six Crew	5	1.6%
AmaiUsagi	4	1.3%
Others (combined)	43	13.5%
Total	**319**	**100%**

Table 18: Agency Distribution

16. Appendix: Key Data Structures

The game state is stored as a single JSON object in localStorage. Below is the current state structure showing all top-level fields and their types. This reference is intended for developers working on the codebase who need to understand the save data format. The structure includes fields for player resources, character collection, studio configuration, gacha state, minigame statistics, stamina, milestone tracking, and pull history.

The top-level game state object contains the following fields: **stars** (number), **starDust** (number), **starFragments** (number), **bondPoints** (number), **gems** (number), **characters** (object keyed by slug, each with level, variant, shards, and ascension status), **studio** (object with level, exp, station assignments and upgrade levels), **gacha** (object with pityCounter and lastPullTimestamp), **stamina** (object with current value and lastRecoveryTimestamp), **minigame** (object with highScore, totalRoundsPlayed, and totalStarsEarned), **milestones** (array of claimed milestone IDs), **pullHistory** (array of pull records with timestamp, banner, and results), **lastSave** (ISO timestamp), **dailyLogin** (object with current streak and lastClaimDate), and **version** (string indicating save data schema version).

The migration system uses the version field to detect when save data needs to be upgraded. When a player loads the game with an older save format, the migration code transforms the data to the latest schema before initialising the game state. This approach ensures backward compatibility and allows the save format to evolve across releases without requiring players to manually reset their progress.

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17. Monetisation Methods (Future Plans)

This section explores potential monetisation strategies for MyVT Gacha Collection. The developer is currently evaluating multiple options and has not committed to any specific approach. The guiding principle for all monetisation considerations is that the game must remain enjoyable and fair for free-to-play players, with any paid elements serving as optional enhancements rather than pay-to-win mechanics. All options presented here are conceptual and subject to change based on community feedback, development capacity, and the evolving needs of the project. Each option is presented with an objective assessment of its revenue potential, development effort, community impact, and implementation timeline.

17.1 Philosophy

The core monetisation philosophy for MyVT Gacha Collection rests on three foundational principles: free-to-play first, no pay-to-win, and community trust. The game was built as a celebration of the Malaysian VTuber community, and any monetisation must enhance rather than undermine that mission. Free-to-play first means that all core gameplay features, including gacha pulls, character progression, studio management, and the minigame, must remain fully accessible without spending any money. No pay-to-win ensures that purchased items provide only cosmetic or quality-of-life benefits, never granting gameplay advantages such as increased pull rates, exclusive characters, or faster resource generation. Community trust means being transparent about monetisation intentions, respecting player time investment, and actively seeking community input before implementing any paid features.

This philosophy reflects the unique position of MyVT Gacha Collection as a community-driven passion project rather than a commercial product. Many players are personal friends of the VTubers featured in the game, and the developer has a responsibility to maintain the goodwill that the project has built within the MyVT community. Any monetisation that feels exploitative or greedy would directly damage the relationships and reputation that the game has been built upon. Therefore, all revenue generation must feel natural, fair, and additive to the player experience.

17.2 Option A - Cosmetic Card Skins

Cosmetic card skins would allow players to customise the visual appearance of their character cards with unique frames, backgrounds, and animation effects. These skins would be purely visual modifications that do not affect any gameplay stat, resource output, or pull rate. Examples include holographic foil frames, animated particle backgrounds, agency-themed borders, seasonal holiday skins, and milestone celebration effects. Skins could be purchased individually using Gems or real currency, or bundled in themed packs. Limited edition skins could be tied to specific events, VTuber birthdays, or community milestones, creating collector value and urgency without gameplay pressure.

The development effort for cosmetic card skins is moderate, estimated at 2-3 weeks for a base implementation including the skin application system, a preview mechanism, and an in-game skin shop interface. Each individual skin asset requires design work, which could be crowdsourced from community artists to reduce development cost while increasing community involvement. Revenue potential is moderate to high depending on player attachment to specific characters, with estimated conversion rates of 5-15% of active players making at least one skin purchase.

Criteria	Assessment
Description	Custom card frames, backgrounds, and animations purchasable with Gems or real currency. Limited editions tied to events or VTuber milestones. Does not affect gameplay.
Revenue Potential	Moderate to High. 5-15% conversion rate. Estimated $0.50-$3.00 ARPU for purchasers.

Dev Effort	Moderate. 2-3 weeks for base system + 1-2 days per skin asset. Can be crowdsourced.
Community Impact	Positive. Players enjoy personalisation. Risk of "paywall" perception if popular skins are expensive.
Implementation Sprint	Sprint 14-15 (post-core features). Depends on skin asset pipeline.

Table 19: Option A - Cosmetic Card Skins Assessment

17.3 Option B - Battle Pass / Season Pass

A seasonal Battle Pass would provide a structured reward track that resets every 30-60 days, offering both a free tier and a premium tier. The free tier would include basic rewards such as small amounts of Stars, Star Dust, and occasional Shards, ensuring that non-paying players still receive value. The premium tier, unlockable for a fixed price (estimated at RM 15-25 or $3-5), would provide enhanced rewards including exclusive cosmetic skins, bonus Gems, rare title badges, and limited-time emotes or profile decorations. Seasons could be themed around MyVT community events, Malaysian cultural celebrations, or VTuber agency milestones to keep content fresh and community-relevant.

The Battle Pass model is well-proven in the gacha gaming space and provides predictable recurring revenue, but it requires significant ongoing development effort to create fresh seasonal content every 30-60 days. Each season needs a new reward track design, new cosmetic assets, themed narrative elements, and balancing adjustments. For a solo developer, maintaining a Battle Pass cadence could become a major development burden that detracts from feature development. Revenue potential is high with estimated conversion rates of 10-20% for premium tier purchases among active players, but the ongoing content creation cost is a significant consideration.

Criteria	Assessment
Description	Free tier with basic rewards; premium tier (RM 15-25) with exclusive cosmetics, bonus Gems, and profile items. 30-60 day seasons tied to MyVT events.
Revenue Potential	High. 10-20% premium conversion. Predictable recurring revenue of $1-3 ARPU per season.
Dev Effort	High. 3-4 weeks initial build + 1-2 weeks content per season. Requires ongoing content pipeline.
Community Impact	Mixed. Engaging for active players but risks FOMO mechanics. Free tier must feel rewarding.

Implementation Sprint	Sprint 15-16. Requires dedicated content creation capacity for sustainability.

Table 20: Option B - Battle Pass Assessment

17.4 Option C - Supporter Membership (Ko-fi / Buy Me a Coffee Integration)

A Supporter Membership model would integrate with Ko-fi or Buy Me a Coffee to offer a low-commitment, community-friendly monetisation channel. Supporters who pledge a monthly amount (e.g., RM 10-30 or $3-8) would receive in-game perks such as a custom supporter nameplate displayed on their profile, an exclusive supporter badge visible in the Collection Gallery, bonus daily login rewards (e.g., +50 Stars per day), early access to new features, and their name listed in a "Supporters" credits section. This model positions the monetisation as a voluntary contribution to support the developer rather than a purchase, which aligns well with the community-driven nature of the project.

The development effort for Ko-fi/Buy Me a Coffee integration is very low, estimated at 3-5 days for the initial implementation. The integration primarily requires a webhook listener to verify membership status and a small UI component to display supporter perks. The verification system can use a simple API call to the Ko-fi/BMC webhook endpoint, storing the supporter status in the player's save data. Revenue potential is low to moderate but highly predictable, with the main value being the establishment of a recurring support channel. Community impact is overwhelmingly positive, as supporters feel directly connected to the developer and the project's continued development.

Criteria	Assessment
Description	Monthly Ko-fi/BMC supporter tier with in-game perks: custom nameplate, supporter badge, bonus daily login rewards, early feature access. Low-commitment, community-friendly.
Revenue Potential	Low to Moderate. Estimated 2-5% conversion. High predictability via recurring pledges.
Dev Effort	Very Low. 3-5 days for webhook integration, status verification, and UI badge display.
Community Impact	Very Positive. Feels like supporting a creator, not buying power. Builds strong dev-community bond.
Implementation Sprint	Sprint 10-11 (can be done alongside other work). Minimal ongoing maintenance.

Table 21: Option C - Supporter Membership Assessment

17.5 Option D - VTuber Collaboration Banners

Collaboration banners would partner with specific MyVT agencies or individual VTubers to create sponsored gacha events. During a collaboration period, a special limited-time banner would feature the collaborating VTubers with enhanced visual effects, unique banner art, and exclusive SR/SSR card variants available only during the event. A portion of any revenue generated from the collaboration banner (e.g., from cosmetic bundle sales or premium pull tickets) would be directed to the featured VTubers or their agencies, creating a win-win scenario that financially supports the creators whose likenesses appear in the game.

This model requires significant coordination effort including negotiations with VTuber agencies, contractual agreements for revenue sharing, creation of bespoke art assets for each collaboration, and careful management of the collaboration calendar. Revenue potential is highly variable depending on the popularity of the featured VTubers and the size of the player base at the time of the collaboration. Community impact is generally very positive as these events celebrate specific members of the MyVT community and can attract new players who are fans of the featured VTubers. However, collaboration banners must be carefully balanced to avoid appearing exploitative or creating pressure to spend money during limited-time events.

Criteria	Assessment
Description	Partner with MyVT agencies for sponsored banners. Exclusive event cards and cosmetic bundles. Revenue share with featured VTubers/agencies.
Revenue Potential	Variable. High during popular collaborations. Depends heavily on player base size at event time.
Dev Effort	High. Requires agency negotiations, legal agreements, bespoke art, and event management.
Community Impact	Very Positive. Celebrates community VTubers. Risk if revenue sharing is perceived as unfair.
Implementation Sprint	Sprint 13-14 (long-term). Requires business development capacity beyond game development.

Table 22: Option D - VTuber Collaboration Banners Assessment

17.6 Option E - Physical Merch Tie-In

Physical merchandise tie-ins would involve selling real-world products such as badges, stickers, keychains, or printed cards at conventions and online stores, with each item containing a unique redeemable code that grants in-game rewards when entered. For example, a convention badge might include a scratch-off code that unlocks an exclusive cosmetic skin or grants bonus pulls. This approach bridges the gap between the digital game and the physical MyVT community, driving offline engagement at conventions, meetups, and events where Malaysian VTubers and their fans gather.

The development effort for the in-game redemption system is moderate (estimated 1-2 weeks), but the physical production and distribution effort is substantial and requires partnerships with merchandising companies or print-on-demand services. Revenue potential is supplementary rather than primary, as the physical margins are thin and distribution is limited. However, the brand visibility and community engagement benefits are significant, and physical merch serves as a powerful marketing tool that can attract new players who discover the game through convention booth visits or friend recommendations.

Criteria	Assessment
Description	Redeemable codes in physical merch (badges, stickers, cards) sold at conventions. Codes unlock exclusive in-game cosmetics or bonus pulls.
Revenue Potential	Low to Moderate. Thin physical margins. Primarily a marketing and engagement channel.
Dev Effort	Moderate for code system (1-2 weeks). High for physical production and distribution.
Community Impact	Very Positive. Drives offline engagement. Creates tangible connection to the game.
Implementation Sprint	Sprint 16+ (long-term). Requires merchandising partnerships and inventory management.

Table 23: Option E - Physical Merch Tie-In Assessment

17.7 Option F - Ad-Supported Free Pulls

Rewarded video ads would provide players with optional opportunities to earn free gacha pulls by watching a short advertisement (approximately 30 seconds). The system would be entirely player-initiated, meaning no forced ads would appear during gameplay. A typical implementation would allow 3 ad-watches per day, each granting 1 free single pull or a small resource bundle. This monetisation method is extremely common in mobile gacha games and is generally well-accepted by players when implemented non-intrusively. The ads would be served through a standard ad network such as Google AdMob, with revenue generated per completed ad view.

The development effort for ad integration is low, estimated at 3-5 days using AdMob's web SDK. However, ad implementation introduces significant user experience considerations including ad loading times, network dependency, ad content quality control, and the potential visual disruption even for optional ads. Revenue per ad view is typically very low (estimated $0.01-$0.05 per completed view), meaning meaningful revenue requires a large and active player base. The community impact is mixed: some players appreciate the option to earn free pulls, while others feel that any advertising compromises the purity of the game experience.

Criteria	Assessment
Description	Optional rewarded video ads (30s). Watch for 1 free pull, 3x daily limit. Player-initiated only via Google AdMob.
Revenue Potential	Low per user ($0.03-0.15/day). Requires large DAU to be meaningful. Scalable with player base.
Dev Effort	Low. 3-5 days for AdMob SDK integration, reward delivery, and daily counter logic.
Community Impact	Mixed. Some appreciate free pulls; others dislike any advertising. Must remain strictly optional.
Implementation Sprint	Sprint 11-12. Quick to implement but requires careful UX design to feel non-intrusive.

Table 24: Option F - Ad-Supported Free Pulls Assessment

17.8 Revenue Projection Model

The following table presents estimated monthly revenue projections for the top three recommended monetisation models across different daily active user (DAU) milestones. These projections are illustrative and based on industry averages for browser-based gacha games with similar player demographics. Actual results will vary significantly based on player engagement, community sentiment, implementation quality, and regional purchasing power. The projections assume the models are implemented individually; combining multiple models would yield additive but diminishing returns due to player spending overlap.

DAU	Ko-fi Supporter (Option C)	Cosmetic Skins (Option A)	Battle Pass (Option B)	Combined Estimate
100	$15-30/mo	$25-50/mo	$30-60/mo	$70-140/mo
1,000	$150-400/mo	$300-750/mo	$400-1,000/mo	$850-2,150/mo

10,000	$1,500-4,000/mo	$3,000-7,500/mo	$4,000-10,000/mo	$8,500-21,500/mo
100,000	$15,000-40,000/mo	$30,000-75,000/mo	$40,000-100,000/mo	$85,000-215,000/mo

Table 25: Monthly Revenue Projections by DAU (Top 3 Models)

17.9 Recommended Strategy

Based on the analysis of all six monetisation options, the recommended strategy follows a phased rollout approach that prioritises low-risk, high-goodwill models before progressing to more complex revenue streams. The phased approach ensures that the developer can validate community reception at each stage and adjust the strategy based on real feedback rather than theoretical projections. The developer is still evaluating these options and reserves the right to modify, combine, or abandon any of the proposed models based on community input and practical considerations.

Phase 1: Community Support Foundation (Sprint 10-11)

Begin with the Ko-fi Supporter Membership (Option C) as the first monetisation implementation. This model requires minimal development effort (3-5 days), has near-zero ongoing maintenance cost, and generates overwhelmingly positive community sentiment. The voluntary nature of Ko-fi support positions the monetisation as a genuine creator-support mechanism rather than a commercial transaction, which aligns perfectly with the project's community-driven identity. Initial perks should be modest: a supporter badge, a custom nameplate, and a small daily login bonus. The Ko-fi page can be set up alongside the in-game integration, providing an immediate revenue channel while the game continues to grow its player base.

Phase 2: Cosmetic Enhancement (Sprint 14-15)

Once the game has a stable player base and the core feature set is complete, introduce Cosmetic Card Skins (Option A) as the second monetisation layer. The initial skin catalogue should include 5-10 skins across different themes (agency-themed, seasonal, milestone-based), priced affordably to encourage broad adoption. Community artists can be commissioned or invited to contribute skin designs, which both reduces development cost and increases community involvement. The skin shop UI should include a preview system that lets players see how skins look before purchasing. Revenue from skins is expected to be higher than Ko-fi supporters but less predictable, making it a good complement to the steady baseline that memberships provide.

Phase 3: Engagement Expansion (Sprint 16+, Long-Term)

The third phase evaluates whether to implement the Battle Pass (Option B) or VTuber Collaboration Banners (Option D) based on the success and reception of Phases 1 and 2. The Battle Pass offers the highest revenue ceiling but demands the most ongoing content creation effort, which may not be sustainable for a solo developer without additional support. Collaboration banners offer a unique community value proposition but require business development capabilities that extend beyond game development. The decision between these two options (or a combination thereof) should be made after evaluating the Phase 1 and Phase 2 results, community growth trajectory, and the developer's available time and resources. Ad-supported free pulls (Option F) and physical merch (Option E) remain viable supplementary options that can be introduced at any phase if the developer chooses to pursue them.

Phase	Option	Timeline	Key Milestone
1	Ko-fi Supporter	Sprint 10-11	Ko-fi page live, in-game badge system
2	Cosmetic Skins	Sprint 14-15	Skin shop with 5-10 initial skins
3	Battle Pass or Collab	Sprint 16+	Decision based on Phase 1-2 results

Table 26: Recommended Phased Monetisation Rollout